Real-Time Transaction Risk & Velocity Enforcement Engine for Point-of-Sale Terminals

Technical Report: RTL-to-GDSII Implementation on IHP SG13G2 (130 nm) with OpenROAD-flow-scripts

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Application domain: FinTech & Digital Payments

Item	Value
Technology / PDK	IHP SG13G2, 130 nm, open PDK
Flow	OpenROAD-flow-scripts (Yosys, OpenROAD, KLayout)
Target clock	20.0 ns (50 MHz)
Throughput / latency	1 record per cycle / 3 cycles
Functional result	4337 results checked, 0 errors
Timing	WNS 0.00 ns, TNS 0.00 ns, setup slack 13.09 ns
Area (logic, excl. fill)	29,321 um^2 (34 % utilization), 2186 cells
Power	1.13 mW total

1. Introduction

Point-of-sale and ATM fraud screening is normally done on backend servers, which adds network latency and depends on connectivity. This project implements a local, deterministic hardware pre-check block. It reads a stream of transaction records and, using only registers and combinational logic, tracks how many transactions and how much spend occurred inside a programmable time window. It then applies four fixed fraud rules and outputs a numeric risk score and a three-way decision: approve, flag for review, or decline.

The objective was to design, verify and physically implement the block from RTL to GDSII using the open-source OpenROAD flow on the IHP SG13G2 130 nm PDK, with no memory macros.

2. Architecture

The design follows the block diagram below. Every block is synchronous and flat; there are no memories. The configuration inputs (window size, count limit, spend limit, anomaly threshold) are quasi-static.

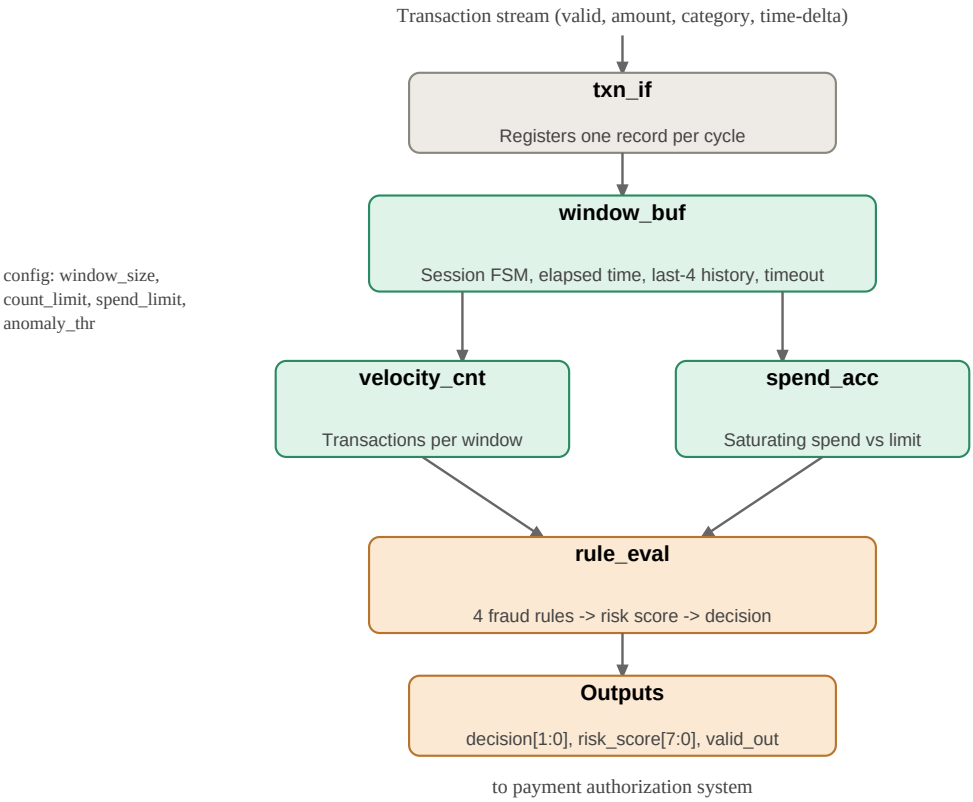


Figure 1. Block diagram of the engine.

Module	Function
txn_if	Registers amount, category and time-delta when valid_in is high.
window_buf	Session FSM (IDLE/ACTIVE); accumulates elapsed time; starts a new window when elapsed time exceeds window_size or no session exists (timeout); keeps the last four (amount, category) records and raises the rapid-repeat flag.
velocity_cnt	8-bit saturating transaction counter, reset to 1 at the start of each window.
spend_acc	24-bit saturating spend accumulator; saturates instead of wrapping so a very large spend can never look small.
rule_eval	Stage A registers the four rule flags; stage B computes the risk score and the decision.
txn_risk_engine	Top level connecting all blocks.

2.1 Fraud rules and decision

Rule	Condition	Weight
Over-limit spend	window spend > spend_limit	100
High velocity	window transaction count > count_limit	60
Rapid repeat	time-delta < 8 and same amount and category as one of the last 4 records	40
Amount anomaly	amount > anomaly_thr	50

The risk score is the sum of the triggered weights, saturated at 255. The decision is **00 (approve)** for a score below 40, **01 (flag for review)** from 40 up to 99, and **11 (decline)** for 100 or more. Because the rule evaluation is pipelined into two stages, a new record can be accepted on every clock cycle.

Note on the FSM: the proposal describes a rule-evaluation FSM. A multi-cycle FSM could not accept one record per cycle, so rule evaluation is implemented as a two-stage pipeline, while the session state

(IDLE/ACTIVE) in window_buf is a genuine FSM.

3. Functional Verification

A self-checking testbench (tb_txn_risk.v) compares every output against an independent behavioural reference model written separately from the RTL. The test sequence covers: normal spending; velocity abuse; spend crossing the limit; rapid repeats; amount anomalies; exact-limit values and window expiry; maximum time-delta and maximum amount; gaps in valid_in; saturation with 300 maximum-value transactions; a reset in the middle of the stream; and a random regression of 4000 records with several different configurations.

Item	Result
Records sent	4340
Results checked	4337
Flushed by deliberate mid-stream reset	3
Errors	0
Verdict	TEST PASSED
Lint (Verilator -Wall)	Clean
Synthesis check (Yosys)	No latches, no warnings

To confirm the testbench actually detects bugs, one rule comparison in the RTL was deliberately changed; the testbench reported failures, then the original RTL was restored.

4. Physical Implementation

The design was taken through the standard ORFS flow: Yosys synthesis, floorplan, tap-cell and power-grid insertion, global and detailed placement, clock tree synthesis, global and detailed routing, fill insertion and GDSII merge. Key configuration: platform ihp-sg13g2, core utilization 30 %, fill enabled, and a 20.0 ns clock with input and output delays of 20 % of the period.

4.1 Results

Metric	Result
Setup timing	WNS 0.00 ns, TNS 0.00 ns; worst setup slack 13.09 ns (met)
Hold timing	worst hold slack 0.27 ns (met)
Clock skew	0.03 ns (setup skew)
Synthesis	1916 cells, chip area 25,922.56 um^2
Final design area (excl. fill)	29,321 um^2, 34 % utilization
Total area incl. fill cells	85,362 um^2 (7717 cells)
Cell count (excl. fill)	2186 cells
Total power	1.13 mW (internal 0.930 mW, switching 0.200 mW, leakage 3.64 uW)

Cell type	Count	Area (um^2)
Sequential cells	197	9,650.79
Multi-input combinational	1627	15,304.46
Inverters	66	359.25
Clock buffers	49	2,222.64
Clock inverters	5	27.22
Timing-repair buffers	242	1,756.34
Fill cells	5531	56,041.37
Total	**7717**	**85,362.08**

Power group	Power	Share
Sequential	0.551 mW	48.6 %
Clock	0.500 mW	44.1 %
Combinational	0.083 mW	7.3 %
Total	**1.13 mW**	**100 %**

Discussion. Timing closes with a very large margin: at 50 MHz the worst setup slack is 13.09 ns, so the design has plenty of headroom and could be clocked well above 50 MHz. Power is dominated by the flip-flops (197 sequential cells, mostly the 4-deep history buffers, the pipeline registers and the accumulators) and by the clock network; combinational logic contributes only about 7 %. Leakage is negligible at 3.64 uW.

5. Layout

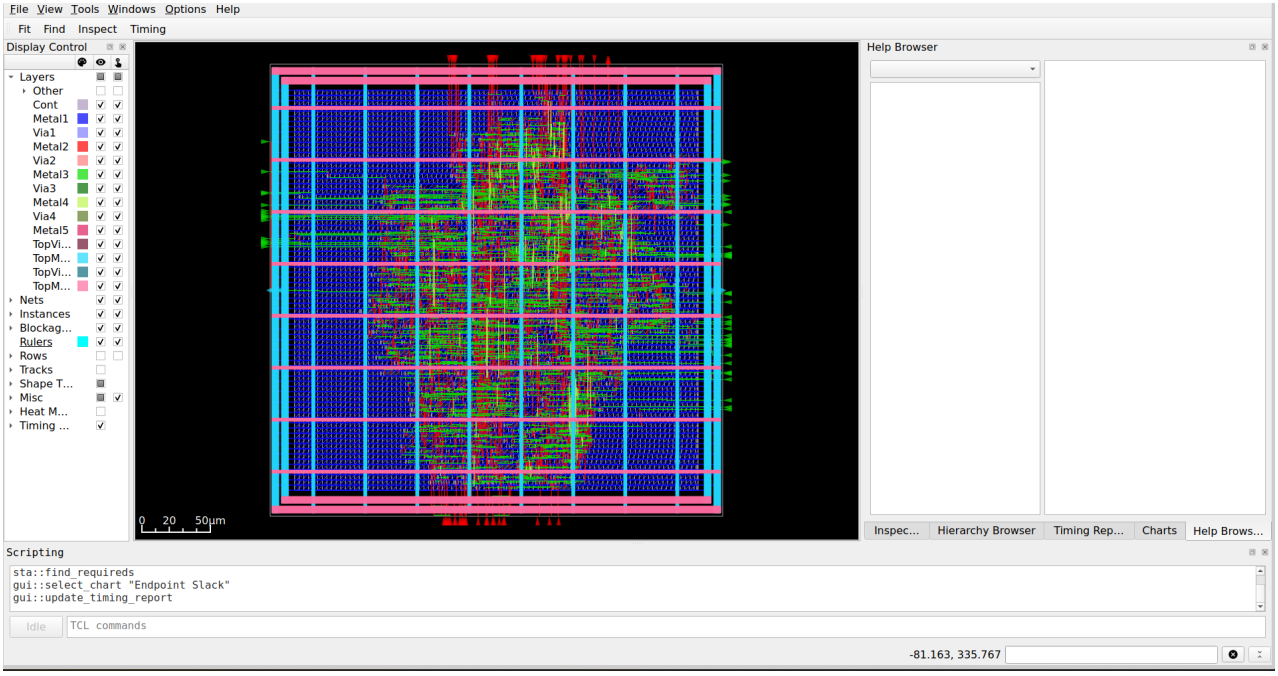


Figure 2. Final placed and routed design in the OpenROAD GUI (power grid, standard cells and signal routing).

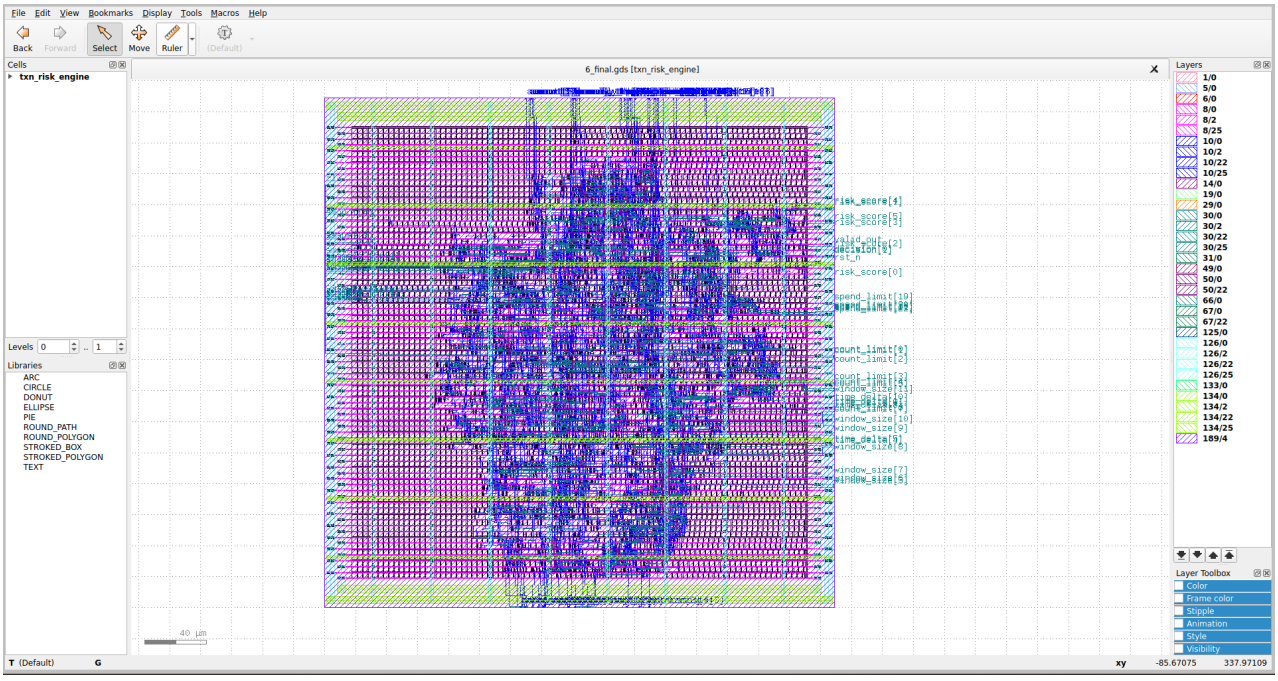


Figure 3. Final GDSII (6_final.gds) opened in KLayout, cell txn_risk_engine, with I/O pin labels.



Figure 4. GDSII view with layer detail.

6. Conclusion and Limitations

The engine was designed, verified and taken to a finished GDSII layout on IHP SG13G2 with a clean timing report and low power. All four fraud rules, the window and timeout handling, and the three-way decision are implemented in fewer than 2200 standard cells, with no memories.

Limitations: timing is evaluated against an ideal clock constraint at 50 MHz; a separate signoff DRC/LVS run was not part of this submission (flow-level checks passed: routing completed, all LEF cells had matching GDS and there were no orphan cells); the design supports a single active session and quasi-static configuration inputs. Possible extensions are multiple concurrent sessions, a true sliding window, and runtime-programmable rule weights.

Appendix: reproducing the results

Simulation: `iverilog -g2012 -o sim tb/tb_txn_risk.v rtl/*.v && vvp sim`
Physical design: copy rtl/*.v to flow/designs/src/txn_risk, copy config.mk and constraint.sdc to flow/designs/ihp-sg13g2/txn_risk, then run `make DESIGN_CONFIG=./designs/ihp-sg13g2/txn_risk/config.mk` from the ORFS flow directory.